TIL Programming: Research Proposal group 22

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1 Dataset and Motivation

In 2024, grid operators warned that overload risks would emerge in Utrecht from 2026 onward (Van Gastel, 2024), while major high voltage grid expansions have slipped from 2029 to 2033 or 2035. In response, the regional grid operator Stedin has requested users to cut consumption during the evening peak between 16:00 and 20:00 (Solar Magazine, 2025). Bus electrification is legally mandated, requiring all new public transport buses to have zero emission from 2025, and the entire fleet by 2030 (Buitelaar, 2016).

A fleet of 250 buses driving 218 km a day consumes roughly 63 MWh per day or 23 GWh a year (Sustainable Bus, 2020), which represents a modest average load of under 3 MW. However, uncontrolled charging upon early evening depot return coincides directly with peak demand (Delmonte et al., 2020). Assuming depot chargers of 60 to 130 kW per bus, 250 buses plugging in at once would draw roughly 15 to 32 MW across a small number of grid connection points. While literature shows that smart charging can reduce peak load by between 7% and 50% (Deb et al., 2022), implementing this in a public transit system introduces strict operational trade-offs. Smart charging for private electric vehicles is often discouraged by driver behaviour, data privacy concerns, and a perceived loss of control (Baumgartner et al., 2023; Delmonte et al., 2020; Günther et al., 2024). In contrast, public bus fleets operate under rigid timetable commitments and fixed vehicle block schedules. Delaying charging past 21:00 is only feasible if minimum operational requirement for subsequent service blocks are maintained.

This project research addresses this conflict by developing an open data Python modelling pipeline to quantify and optimize the temporal and spatial load profiles of bus charging in Utrecht, the heart of the Netherlands. By integrating open static GTFS timetable data with fleet power characteristics, the model evaluates how much peak demand can be shifted out of the 16:00–20:00 window under explicit constraints of the GTFS and operating schedule limits.

2 Research Questions

2.1 Main Research Question

To what extent does the charging demand of electric buses in Utrecht contribute to local electricity grid peaks, and how can charging besides these peak moments mitigate this impact?

2.2 Subquestions

1. **What is the hourly electricity demand of the electric bus fleet in Utrecht, derived by filtering regional route and timetable data from open GTFS datasets?**
2. **During which hours of the day does the combined load of regular electricity use and bus charging result in peak capacity pressure on the Utrecht grid?**
3. **What are the daily electricity consumption profiles and peak load periods in the Utrecht region based on open energy datasets?**
4. **How effectively does shifting bus charging from daytime operational hours to overnight off-peak periods reduce peak electricity load in Utrecht?**

References

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